

Software Requirements Specification

Amar Shohor

A civic problem reporting and accountability platform

One city, one map

FIELD	VALUE
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Standard	Structure adapted from IEEE Std 830-1998

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1 Introduction

1.1 Purpose

This document specifies the requirements for **Amar Shohor**, a web platform on which citizens report physical problems in their city. Repeated reports of the same thing are collapsed into a single verified problem, which is then ranked, assigned to a municipal department and tracked in public until it is resolved. It is written for the project supervisors and examiners, and for the five developers listed on the cover, who own the areas it describes.

The document records the system **as built**. Where a capability is specified but not yet implemented it is marked as such rather than written in the present tense; section 10 collects those gaps in one place.

1.2 Scope

The platform covers the full lifecycle of a civic problem: submission by a citizen, automated deduplication against existing problems, explainable priority scoring, community verification, departmental triage and assignment, resolution with evidence, and public reporting of the resulting statistics.

IN SCOPE	OUT OF SCOPE
Citizen reporting with photograph and location	Payment, billing or procurement workflows
Automatic deduplication of repeated reports	Dispatch, scheduling or crew management
Explainable priority ranking	Predictive forecasting of future problems
Departmental triage with a service-level clock	SMS or push notification delivery
Public map, timeline and accountability dashboard	Native mobile applications

1.3 Definitions and acronyms

TERM	MEANING
Report	One submission by one citizen: a photograph, a location, a category and an optional description.
Issue	One verified physical problem. Several reports of the same thing collapse into one issue.
Deduplication	The scored decision that attaches an incoming report to an existing issue rather than creating a new one.
Priority score	A value from 0 to 100 derived from five weighted factors, published together with its breakdown.
Status event	An append-only record of one lifecycle change. Serves as both public timeline and audit log.
Proof of fix	A photograph attached when a problem is closed, showing the completed work.
Sign-off	The reporting citizen accepting, or rejecting, an authority claim that a problem is resolved.
SLA	Service level agreement: the time budget a department has to resolve a category of problem.
Ward	An administrative subdivision of the city, stored as a polygon and carrying a population density.
Staff code	A time-limited, use-limited invitation allowing an account to be created with authority rights.

1.4 References

ID	REFERENCE
R1	IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
R2	Project repository: https://github.com/MrTimonM/amar-shohor-
R3	Deployed system: https://amar-shohor.vercel.app
R4	IMPLEMENTATION_PLAN.md, the fourteen-phase build plan
R5	PRODUCT.md, product positioning, principles and evidence register
R6	RFC 2606, reserved top-level domains, used for every seeded demonstration account
R7	WCAG 2.1 Level AA, the accessibility conformance target

1.5 Document overview

Section 2 describes the product and its users in prose. Section 3 gives the architecture and section 4 the data model. Section 5 contains the numbered functional requirements, which are the testable core of this document. Sections 6 and 7 cover interfaces and quality attributes. Sections 8 to 10 record how the system is deployed, how it was verified, and what it deliberately does not do. Section 11 indexes every requirement identifier.

2 Overall description

2.1 Product perspective

Amar Shohor is a self-contained three-tier web system with no dependency on an existing municipal IT estate. It replaces the prevailing alternative, which is a telephone line, a paper register or a social-media post, with a shared public record.

The central claim

Fifty people reporting the same pothole produce **one problem with a report count**, not fifty tickets. Every surface in the product is built to make that collapse visible, because it is the difference between an accountability tool and a complaint box.

2.2 Product functions

#	FUNCTION	PRIMARY ACTOR
F1	Submit a problem report with photograph, location and category	Citizen
F2	Collapse duplicate reports into a single verified problem	System
F3	Rank problems by an auditable priority score	System
F4	Confirm or dispute a problem while near it	Citizen
F5	Review deduplication decisions the system was not confident about	Moderator
F6	Triage, assign and progress problems within a department	Authority staffer

#	FUNCTION	PRIMARY ACTOR
F7	Close a problem with a note and an optional proof-of-fix photograph	Authority staffer
F8	Accept or reject a claimed resolution	Reporting citizen
F9	Issue and withdraw staff access	Administrator
F10	Read the live map, any problem timeline and the accountability dashboard	Public observer

2.3 User classes and characteristics

CLASS	CHARACTERISTICS	PRIVILEGE
Public observer	No account. Opens the map or dashboard to see what is broken nearby and whether anything is being done. This is the primary user: where a design decision helps a signed-in role at this user expense, this user wins.	Read only
Citizen	On the street, on a mid-range phone, often on a poor connection. Needs to file a report in about a minute.	Report, verify
Verifier	A citizen with an elevated trust score, whose confirmations carry proportionally more weight.	Weighted verify
Authority staffer	At a desk, working one department queue. Needs priority order, assignment and a visible clock.	Own department
Administrator	Runs the deployment. Issues and withdraws staff access across all departments.	Unrestricted

2.4 Operating environment

TIER	ENVIRONMENT
Client	Any modern browser, mobile or desktop. Designed against a mid-range Android device on an intermittent connection.
Server	Node.js 20 or later. Runs either as a long-lived process or as a serverless function; the application code is identical in both.
Database	MongoDB 7, or MongoDB Atlas, with 2dsphere geospatial indexing.
Intelligence	Python 3.12 with FastAPI. Optional: the platform runs correctly with it switched off.

2.5 Design and implementation constraints

- **Reading must never require an account.** The public map and dashboard are open to anonymous visitors on any device.
- **Bengali is the default language** and English the alternate. Bengali typography requires a looser line height than the Latin scale.
- **Nothing automated may be unexplained.** A merge, a rank, a status change and a model prediction must each show what produced them.
- **The citizen is never blocked.** Submission succeeds before enrichment, offline precedes online, and the citizen own category overrides the model.
- **Unbuilt capability is stated on the page** rather than mocked.

- **Configuration comes only from the environment.** No URL, key or bucket is hardcoded, which is what makes redeployment a configuration change rather than a rewrite.

2.6 Assumptions and dependencies

ID	ASSUMPTION OR DEPENDENCY	CONSEQUENCE IF IT DOES NOT HOLD
A1	The citizen device can supply a photograph and a location fix.	A report cannot be filed. Both are mandatory inputs.
A2	Ward boundaries and population densities are available as polygons.	The density factor of the priority score cannot be computed.
A3	A municipal department will act on an assigned problem.	The record stays public but resolution statistics remain at zero.
A4	MongoDB is reachable from the API.	The API refuses to start and reports the cause rather than failing obscurely.
A5	The intelligence service may be unavailable at any time.	Handled by design: enrichment is asynchronous and the match score renormalises over the factors it does have.

3 System architecture

3.1 Component view

COMPONENT	RESPONSIBILITY	TECHNOLOGY
Shared domain package	The single source of truth for categories, the lifecycle, the priority algorithm and every request and response shape. Both the API and the web client import it, so a field cannot drift between them.	TypeScript, Zod
Web client	Public map, report flow, problem pages, triage queue, dashboard and admin console.	React 18, Vite 6, Leaflet
API	Authentication, authorisation, the deduplication engine, the media pipeline and all persistence.	Express 4, Mongoose 8
Intelligence service	Photograph classification, severity estimation, relevance checking and image embedding.	FastAPI, Pillow, NumPy
Database	Wards, users, reports, issues, status events, verifications, staff codes and photograph bytes.	MongoDB

3.2 Request lifecycle

The path a report takes, from the citizen pressing submit to the problem appearing on the map:

STEP	WHAT HAPPENS
1	The client compresses the photograph and uploads it, receiving an opaque photo identifier. A URL is never accepted from the client.
2	The client submits the report referencing that identifier. The server validates it against the shared schema and writes the report.

STEP	WHAT HAPPENS
3	The response returns immediately. Enrichment is queued rather than awaited, so a slow model never delays a citizen.
4	The enrichment job requests classification, severity and an embedding from the intelligence service, tolerating its absence.
5	The deduplication engine finds candidate issues within the configured radius and time window, then scores each one.
6	At or above the auto-merge threshold the report joins an existing issue. Between the thresholds it waits for a moderator. Below, it opens a new issue.
7	The priority score is recomputed for the affected issue and stored with its full factor breakdown.

Why the response comes first

Inference and deduplication run after the report is safely written. A citizen standing next to a flooded street should never wait on a model, and a failed model must never cost a report.

3.3 Technology stack

LAYER	CHOICE	REASON
Language	TypeScript across client, server and shared package	One vocabulary and one set of types across the whole wire.
Database	MongoDB with 2dsphere indexes	The domain is document-shaped, and every map query and deduplication candidate search is geospatial.
Map	Leaflet driven directly rather than through a wrapper	The map holds imperative state that a declarative wrapper only re-wraps.
Styling	Plain CSS over a token layer	Light, dark and system themes resolve from one set of variables rather than three stylesheets.
Passwords	bcrypt from the Node standard library	bcrypt and argon2 are native modules that fail to build on exactly the machines a demonstration runs on.

4 Data requirements

4.1 Entity model

The distinction the whole product rests on is between a **report** and an **issue**. They are separate collections, not one collection with a flag.

RELATIONSHIP	CARDINALITY	MEANING
Report to Issue	many to one	Several citizen submissions collapse into one physical problem.
Issue to Status event	one to many	The append-only public timeline of that problem.
Issue to Verification	one to many	At most one confirmation or dispute per person per issue.
User to Report	one to many	A citizen keeps ownership of everything they submit.
Ward to Issue	one to many	Stamped on write from the ward polygon containing the location.

4.2 Collections and key fields

COLLECTION	KEY FIELDS
users	email (unique), passwordHash (never selected by default), name, role, department, trust, reportCount
reports	ref, category, severity, location, photos, reporterId, wardId, ai (model, confidence, embedding), issueId, mergeConfidence, mergeDecision
issues	ref, title, category, status, location (centroid), reportIds, reportCount, photos, proofPhotos, priorityScore, priorityBand, priorityFactors, embedding, assigneId, department, slaDueAt, citizenSignedOffAt
statusevents	issueId, status, from, note, actorId, actorName, actorRole, proofPhotos, at
verifications	issueId, userId, vote, weight (trust at vote time), distanceM
wards	name, cityCorporation, densityPerKm2, center, boundary (polygon)
staffcodes	code, department, label, expiresAt, maxUses, useCount, revokedAt
uploads	photo, ownerId, claimedAt, expiresAt (self-expiring)

4.3 Indexes and integrity

INDEX OR RULE	PURPOSE
2dsphere on report and issue location	Map viewport queries and deduplication candidate search. Without it a bounded lookup degrades to a full scan.
status with descending priority score	The public map ordering and the triage queue ordering.
department, status, SLA due date	The departmental queue, in the order its fields are filtered.
Unique issue and user pair on verifications	Enforces one vote per person per problem at the database level.
Time-to-live on uploads	An unclaimed photograph removes itself rather than accumulating.
Append-only status events	Update and delete operations are refused by the model layer, so the audit trail cannot be rewritten even by mistake.

Integrity is enforced, not documented

The append-only rule is implemented as a refusal on six mutating operations rather than as a convention. An accountability record whose history can be edited accounts for nothing.

5 Functional requirements

Each requirement has a stable identifier, a priority, and a statement written so that it can be tested. **M** marks mandatory, **S** should-have, **C** could-have.

5.1 Account management

ID	PR	REQUIREMENT
FR-1.1	M	A visitor shall be able to create an account with a name, an email address and a password of at least eight characters.
FR-1.2	M	An account shall be usable immediately on creation. The system shall not require email confirmation.
FR-1.3	M	Passwords shall be stored only as a script hash, with the cost parameter carried inside the stored value so that it can be raised later without invalidating existing hashes.
FR-1.4	M	The system shall return an identical response for an unknown email address and for a wrong password, so that the form cannot be used to discover which addresses are registered.
FR-1.5	M	A sign-up shall create an authority account only when it carries both a department and a valid staff code. A sign-up carrying a department without a code shall be refused.
FR-1.6	M	Sign-in shall return a signed session token with a bounded lifetime.
FR-1.7	M	Sign-up and sign-in shall be rate limited per email address.
FR-1.8	S	Reading the map, a problem page or the dashboard shall never require an account.

5.2 Problem reporting

ID	PR	REQUIREMENT
FR-2.1	M	A signed-in citizen shall be able to submit a report consisting of at least one photograph, a location, a category from the published set of eight, and an optional description.
FR-2.2	M	The client shall compress photographs before upload; the server shall reject anything over twelve megabytes.
FR-2.3	M	The server shall verify the file type from its magic bytes rather than from the declared content type.
FR-2.4	M	A report shall reference photographs by opaque identifier. The server shall never accept a photograph URL from a client.
FR-2.5	M	Submission shall return before enrichment runs. A slow or absent intelligence service shall not delay or fail a submission.
FR-2.6	M	A report submitted without a connection shall be stored on the device and sent automatically when connectivity returns.
FR-2.7	S	Reports still waiting to send shall be listed to their author rather than hidden.
FR-2.8	S	Before submitting, the citizen shall be shown existing problems nearby so that a duplicate can be confirmed instead of filed.
FR-2.9	M	Each report shall be stamped with the ward containing its location at the time it is written.

5.3 Deduplication

ID	PR	REQUIREMENT
FR-3.1	M	On enrichment, the system shall find candidate issues within a configurable radius and time window, excluding resolved and rejected ones.
FR-3.2	M	The system shall score each candidate over five factors: photograph similarity, distance, category agreement, time proximity and description overlap.

ID	PR	REQUIREMENT
FR-3.3	M	The score shall be renormalised over only those factors for which data was available, so that a missing photograph embedding does not depress every score.
FR-3.4	M	At or above the auto-merge threshold the report shall be attached to the matching issue; between the two thresholds it shall be held for human review; below the review threshold a new issue shall be created.
FR-3.5	M	Both thresholds shall be configurable without code changes.
FR-3.6	M	The match score and its per-factor breakdown shall be stored on the report.
FR-3.7	M	A moderator shall be able to approve, reject or redirect any held decision, and every merge shall be reversible.
FR-3.8	S	A problem page shall display the reports merged into it, with their photographs and the count.

Why two thresholds rather than one

The two failure modes are not symmetric. A missed merge produces a slightly noisy map. A wrong merge **hides a real problem**: the citizen report disappears into someone else issue and they are never told. The uncertain band therefore goes to a person rather than to a guess.

5.4 Priority scoring

ID	PR	REQUIREMENT
FR-4.1	M	Every issue shall carry a priority score from 0 to 100, computed from five weighted factors whose weights total exactly 100.
FR-4.2	M	The factors shall be severity (32), number of reports (24), days open (18), affected population density (14) and hazard classification (12).
FR-4.3	M	The report-count factor shall use a logarithmic curve, so that the step from one report to three counts for more than the step from ten to twelve.
FR-4.4	M	Every API response carrying a score shall also carry each factor with its normalised value, weight, points contributed and a plain-language reason.
FR-4.5	M	The client shall render the stored breakdown and shall never recompute a score.
FR-4.6	M	The score shall be recomputed whenever a report is merged in or the status changes.
FR-4.7	S	Scores shall be banded as critical, high, medium or low for display.

5.5 Community verification

ID	PR	REQUIREMENT
FR-5.1	M	A signed-in citizen shall be able to confirm or dispute a problem.
FR-5.2	M	A vote shall be accepted only from a device within a configurable distance of the problem.
FR-5.3	M	Each vote shall be weighted by the voter trust score, clamped to a bounded range so that a new account cannot carry a problem alone.
FR-5.4	M	The weight shall be recorded as it stood at the time of voting, so that history stays reproducible when trust later changes.

ID	PR	REQUIREMENT
FR-5.5	M	A problem shall become verified once weighted confirmations reach the published threshold.
FR-5.6	M	A person shall vote at most once per problem, enforced by a unique database index.
FR-5.7	S	Confirming problems that are later resolved shall raise the voter trust score.

5.6 Authority workspace

ID	PR	REQUIREMENT
FR-6.1	M	An authority account shall see a queue scoped to its own department, ordered by priority score.
FR-6.2	M	The queue shall be filterable by assignment state and by overdue status.
FR-6.3	M	Only an administrator, the owning department, or the person a problem is assigned to shall be able to change its status.
FR-6.4	M	Only an administrator shall be able to assign a problem to a different department.
FR-6.5	M	Every status change shall carry a note of at least three characters, published on the timeline.
FR-6.6	M	Only transitions permitted by the published lifecycle table shall be accepted; anything else shall be refused with a conflict response.
FR-6.7	M	Assignment shall start a service-level clock derived from the category time budget.
FR-6.8	S	Closing a problem shall accept an optional proof-of-fix photograph, which must belong to the person closing it.
FR-6.9	M	The citizen who reported a problem shall be able to accept or reject a claimed resolution.
FR-6.10	M	Rejecting a resolution shall return the problem to assigned and clear its resolution timestamp.

5.7 Administration

ID	PR	REQUIREMENT
FR-7.1	M	An administrator shall be able to issue a staff code, optionally scoped to a department, with an expiry and a maximum number of uses.
FR-7.2	M	A code scoped to a department shall pin any account created with it to that department.
FR-7.3	M	An administrator shall be able to withdraw a code, taking effect immediately.
FR-7.4	M	A withdrawn, expired or exhausted code shall remain listed as a record of what was issued and by whom.
FR-7.5	M	A code that has created at least one account shall not be erasable, only withdrawable.
FR-7.6	M	An administrator shall be able to list every account holding elevated access, with the number of problems currently assigned to each.
FR-7.7	M	Revoking access shall demote the account rather than delete it, preserving authorship of the events it wrote.
FR-7.8	M	Deleting an account shall be refused while problems remain assigned to it.
FR-7.9	M	An administrator shall not be able to remove their own administrator role or account.

ID	PR	REQUIREMENT
FR-7.10	M	Every administration route shall be enforced by server-side middleware, not by hiding controls in the interface.

5.8 Public map and dashboard

ID	PR	REQUIREMENT
FR-8.1	M	The map shall request only problems within the current viewport.
FR-8.2	M	Nearby markers shall cluster, expanding as the viewer zooms in.
FR-8.3	M	Filters shall be reflected in the address bar so that a filtered view can be shared as a link.
FR-8.4	M	A list view carrying the same data as the map shall be available as a first-class equivalent, not as a fallback.
FR-8.5	M	Resolved problems shall be hidden from the default map view and shall remain reachable through a filter, a direct link and the dashboard.
FR-8.6	M	A problem page shall show the merged reports, the priority breakdown, the verification state, the full timeline and any proof of fix.
FR-8.7	M	The dashboard shall publish resolution rate, median time to resolution, open backlog and service-level breaches, by ward and by department.
FR-8.8	S	Any statistic the system cannot yet produce shall be stated as absent on the page rather than filled with invented numbers.

6 External interface requirements

6.1 User interfaces

SCREEN	PURPOSE	ACCESS
Home	States what the map is and routes the two kinds of account holder to the right door.	Public
Live map	Clustered problems over the city, with filters and a paired list view.	Public
Problem page	Merged reports, priority breakdown, verification, timeline, proof of fix.	Public
Report	Photograph, location, category and description, in about a minute.	Citizen
My reports	Everything the citizen has submitted, including anything still unsent.	Citizen
Dashboard	Resolution rate, median time to fix, backlog and breaches by ward and department.	Public
Triage queue	The department workload in priority order, with the service-level clock.	Authority
Merge review	Deduplication decisions held for a human, with the score that produced the hold.	Authority
Staff and access	Issue and withdraw staff codes; list and demote accounts holding access.	Administrator

6.2 Application programming interface

A JSON over HTTP interface. Reads are anonymous; writes require a bearer token. Request and response shapes are defined once in the shared package and validated on the server.

METHOD AND PATH	PURPOSE	ACCESS
POST /v1/auth/signup	Create an account, optionally with a staff code	Public
POST /v1/auth/login	Exchange credentials for a session token	Public
GET /v1/auth/me	The signed-in account	Token
GET /v1/issues	Problems, filtered by viewport, category, status, ward or band	Public
GET /v1/issues/:id	One problem with reports, timeline and breakdown	Public
POST /v1/issues/:id/verify	Confirm or dispute, gated on proximity	Citizen
POST /v1/media/photos	Upload photographs, returning opaque identifiers	Citizen
POST /v1/reports	Submit a report	Citizen
GET /v1/reports/mine	The caller own reports	Citizen
GET /v1/authority/queue	The department triage queue	Authority
POST /v1/authority/issues/:id/status	Move a problem through its lifecycle	Authority
POST /v1/authority/issues/:id/signoff	Accept or reject a claimed resolution	Reporter
GET, POST /v1/admin/staff-codes	List or issue staff codes	Administrator
DELETE /v1/admin/staff-codes/:id	Withdraw or erase a staff code	Administrator
GET /v1/admin/staff	Accounts holding elevated access	Administrator
GET /v1/stats/dashboard	Aggregate accountability statistics	Public
GET /health	Liveness, storage driver, queue depth, model availability	Public

6.3 Software interfaces

INTERFACE	CONTRACT
MongoDB	Accessed only through Mongoose models. The connection string is supplied by the environment.
Intelligence service	HTTP with a bounded timeout. Every call has a defined fallback; the platform is fully functional without it.
Object storage	A storage adapter interface with three drivers: local disk, database, and object store. Swapping them is a configuration change.
Map tiles	OpenStreetMap raster tiles, with the basemap changing to match the active theme.

7 Non-functional requirements

7.1 Performance

ID	REQUIREMENT
NFR-1.1	A map viewport query shall return only problems inside the requested bounds, so response size does not grow with the size of the database.
NFR-1.2	Deduplication candidate generation shall use an indexed geospatial query capped at twenty-five candidates, so scoring cost stays flat as the problem count grows.
NFR-1.3	Report submission shall respond without waiting for classification or deduplication.
NFR-1.4	Photographs shall be resized and re-encoded server side, and a separate thumbnail served to list and map views.
NFR-1.5	Database connections shall be reused across serverless invocations rather than opened per request.

7.2 Security

ID	REQUIREMENT
NFR-2.1	Passwords shall never be stored or logged in a recoverable form, and the hash field shall be excluded from queries by default.
NFR-2.2	Authentication and role checks shall be enforced in server middleware. Hiding a control in the interface is not an access control.
NFR-2.3	Authorisation shall be scoped by department as well as by role, so that one department cannot close another work.
NFR-2.4	Uploaded files shall be validated by magic bytes, size limited, and re-encoded, which strips embedded metadata from what is served.
NFR-2.5	Location metadata extracted from a photograph shall be retained for integrity checking and shall never appear in an API response.
NFR-2.6	Authentication endpoints shall be rate limited per identifier.
NFR-2.7	Secrets shall be supplied by the environment and shall never be committed to the repository.
NFR-2.8	The system shall refuse to start in production with a development signing secret.

7.3 Reliability and availability

ID	REQUIREMENT
NFR-3.1	Loss of the intelligence service shall degrade enrichment only; reporting, reading and triage shall continue unaffected.
NFR-3.2	Loss of connectivity on the client shall queue a report on the device rather than discard it.
NFR-3.3	An unreachable database shall produce a readable startup error rather than an obscure runtime failure.
NFR-3.4	A long-running deployment shall finish in-flight requests before exiting, so that a restart does not drop a report mid-submission.
NFR-3.5	The audit trail shall be append-only and shall reject modification at the model layer.

7.4 Usability and accessibility

ID	REQUIREMENT
NFR-4.1	The interface shall meet WCAG 2.1 Level AA contrast in both the light and the dark theme.
NFR-4.2	Light, dark and system themes shall resolve from one token layer, and the stored preference shall be applied before first paint.
NFR-4.3	Interactive targets shall be at least 44 by 44 pixels, with primary actions within thumb reach on a phone.
NFR-4.4	The interface shall be fully usable by keyboard, with a visible focus indicator throughout.
NFR-4.5	Reduced-motion preferences shall be respected.
NFR-4.6	Every screen shall be complete in both Bengali and English, with Bengali as the default.
NFR-4.7	Chart colours shall be validated for colour-vision deficiency in both themes.
NFR-4.8	Filing a report shall take about one minute on a mid-range phone.

An accessibility decision that changed the design

The obvious green and amber pairing for the trend chart failed deuteranopia separation when run through a colour-vision validator. The chart is blue and orange as a result. That is a measured outcome, not a stylistic preference.

7.5 Maintainability and portability

ID	REQUIREMENT
NFR-5.1	Domain rules, the scoring algorithm and every wire shape shall live in one shared package imported by both client and server.
NFR-5.2	Storage, inference and queueing shall sit behind interfaces so that each can be replaced by configuration.
NFR-5.3	All configuration shall come from the environment and shall be validated at startup, failing with a readable message.
NFR-5.4	The same application code shall run as a long-lived server and as a serverless function.
NFR-5.5	The repository shall carry a continuous integration pipeline that type-checks and builds on every change.

8 Deployment

The system is deployed and publicly reachable. The same codebase runs locally as a long-lived server and in production as a serverless function; which one it is on is decided by configuration rather than by a separate build.

ASPECT	PRODUCTION	LOCAL DEVELOPMENT
Web client	Static build served from a global edge network	Vite development server on port 5173
API	Serverless function, requests routed to it by rewrite rules	Long-lived Node process on port 4000
Database	MongoDB Atlas	Atlas or a local MongoDB
Photograph storage	Database-backed driver, because serverless hosting has no persistent disk	Local disk under the API working directory

ASPECT	PRODUCTION	LOCAL DEVELOPMENT
Background work	Executed inline, since a function is frozen once it responds	In-process queue with bounded concurrency
Intelligence service	Disabled	Optional FastAPI process on port 8000

One codebase, two runtimes

A serverless function is frozen the moment it responds, so work scheduled for after the response would never run, and it would fail **silently**: reports written but never deduplicated. The queue therefore detects the absence of a long-lived process and runs the job inline. Submission gets slower; it stays correct.

9 Verification

The following were confirmed by executing the system, not by inspection.

AREA	VERIFIED	RESULT
Scoring	Factor points sum to the reported score; weights total exactly 100	Pass
Geometry	Haversine distance returns 99.9 m for a computed 100 m offset	Pass
Lifecycle	Legal transitions accepted; illegal transitions refused with a conflict	Pass
Vectors	Cosine similarity returns 1.0 identical, 0 orthogonal, 0 on length mismatch rather than failing	Pass
Authentication	Sign-up, sign-in, wrong password, unknown address, duplicate address, short password	Pass
Authorisation	A staffer of another department refused; owning department and assignee accepted; administrator unrestricted	Pass
Staff codes	Issue, consume, count uses, withdraw, refuse a withdrawn code, refuse erasing a used code	Pass
Self-protection	An administrator cannot remove their own role or account	Pass
Rendering	Home, sign-in, map and admin console rendered in a headless browser with zero console errors	Pass
Build	Type-check of API and client, and a production client build	Pass
Deployment	Health, problem listing, sign-in and static routes answered on the live URL	Pass
Dark theme	Checked structurally only; not yet compared side by side in a browser	Partial
Automated tests	No automated test suite exists; the checks above were run by hand	Absent

10 Limitations and future work

Stated plainly, because a gap that is declared is worth more than one that is discovered. Nothing below is presented anywhere in the product as finished.

LIMITATION	CURRENT STATE	PLANNED RESOLUTION
Vision models are not trained	Colour, texture and edge heuristics. The registry marks each model trained or untrained; the classifier caps its own confidence at 0.62 and the citizen category always wins.	A labelled Bangladeshi street-imagery dataset, then a fine-tuned vision transformer. Every citizen override is already logged as training signal.
Embeddings are perceptual hashes	A discrete cosine transform hash. Genuinely effective for near-duplicates of the same viewpoint.	A learned embedding, so that photographs from different angles also match.
Vector search runs in process	Cosine similarity over the candidates the geospatial query returns.	Managed vector search. One function is the only call site.
No automated test suite	Verified by hand, as recorded in section 9.	Unit tests for scoring and matching; integration tests for the authorisation paths.
Predictive hotspots absent	The dashboard states this on the page instead of showing invented numbers.	Requires a full monsoon of history before a forecast carries meaning.
No SMS or push delivery	Both are gateway integrations that do not exist.	A notification service. The contact field is already on the user record.
Staff codes are shared secrets	Time limited, use limited and revocable, but not tied to one person.	Per-person invitations bound to an email address.
Photographs stored in the database	Works, and avoids a second cloud account for a demonstration.	An object store behind a content delivery network. The storage adapter is the seam.

11 Team contributions

11.1 Division of work

The project was divided into five areas along module boundaries, so that each member owns a coherent part of the system rather than a scattering of files. Ownership means the member wrote or maintains that area, is the reviewer for changes to it, and answers for it under examination.

MEMBER	ID	GITHUB	AREA OWNED	FILES
Mahedi Hasan Sorol	0112310359	MrTimonM	Domain model, priority scoring, deduplication engine	16
Antor Ghosh	0112330131	antorghSD	Backend API, authentication, authorisation, administration	20
Tanjid Ahmad Anik	0112230651	tanjidahmad	Web platform, design system, theming, accessibility	21
Sampriti Biswas Joya	0112230018	joyabiswas	Application screens, map, data visualisation	13
Meherin Sohani	0112330420	meherin20	Intelligence service, CI pipeline, QA, documentation	14

Work reaches the shared branch through pull requests. The repository carries a code-owners file mapping each directory to its owner, so the correct reviewer is requested automatically. Three files are deliberately shared between areas and are the only expected source of conflict: the client API wrapper, the application router, and the shared request schemas.

11.2 Mahedi Hasan Sorol

Domain model, priority scoring and the deduplication engine — the rules the rest of the system is built on, and the repository foundation.

- The shared domain package: eight problem categories with their owning department, hazard flag and service-level budget; the six-state lifecycle and the table of legally permitted transitions; the role and department vocabulary.
- The explainable priority algorithm: five weighted factors totalling exactly 100, a logarithmic curve on report count, and a return type of factors rather than a bare number so that every score can be published with its own reasoning.
- The data model: the separation of report from issue, the append-only status event with modification refused at the model layer, and the geospatial and compound indexes every query depends on.
- The deduplication engine: candidate generation by indexed geospatial query, five-factor match scoring, and renormalisation over the factors that actually carried data.
- Repository foundation: workspace configuration, ignore rules and line-ending normalisation across five machines.
- This requirements specification.

11.3 Antor Ghosh

The backend API, authentication, authorisation and the administration console — every route, and every rule about who may do what.

- The complete HTTP surface: authentication, reports, issues, media, authority workspace, administration and statistics.
- Password authentication using script from the standard library, with the cost parameter stored alongside each hash, constant-time comparison, and an identical response for an unknown address and a wrong password.
- Authorisation in middleware rather than in the interface, scoped by department as well as by role: an administrator, the owning department, or the assignee may act on a problem, and nobody else.
- The staff code lifecycle: issue, scope to a department, limit by expiry and use count, consume on sign-up, and withdraw.
- The media pipeline: file type verified from magic bytes, size limits, server-side resizing and re-encoding, derivative thumbnails, and metadata retained for integrity checking but never served.
- The storage adapter and its drivers, and the seed and staff provisioning scripts.

11.4 Tanjid Ahmad Anik

The web platform, design system and accessibility — the foundation every screen is built on.

- The client application shell: build configuration, routing, navigation for desktop and mobile, and the error surface that reports the first failure without destroying the page reporting it.
- The design token layer: the complete light palette defined once, only the changed tokens redefined for dark, and the stored preference applied before first paint so that no theme flashes.
- The reusable component kit: cards, pills, meters, banners, empty states, segmented controls, sheets, tables and the icon set, all built to the WCAG AA contrast floor in both themes.
- Client libraries: the typed API wrapper, the authentication context, and the bilingual interface context with Bengali as the default language.

- The offline queue: a report filed without a connection is stored on the device and sent automatically when connectivity returns.
- Accessibility: visible focus, keyboard reachability, reduced-motion support and minimum touch target sizes.

11.5 Sampriti Biswas Joya

Every screen a user actually looks at, including the map and the data visualisation.

- Ten screens: home, live map, problem detail, report submission, my reports, triage queue, merge review, administration, accountability dashboard and sign-in.
- The map canvas: marker clustering, viewport-bounded querying so payload does not grow with the database, theme-aware base layers, and a list view carrying the same data as a first-class equivalent rather than a fallback.
- The merge proof presentation, which puts the collapse of several reports into one problem on screen with the contributing photographs.
- The priority breakdown panel, which renders the factors exactly as the server computed them and never recalculates a score.
- The public timeline, verification panel and proof-of-fix presentation on the problem page.
- The dashboard charts, with a palette validated for colour-vision deficiency in both themes.

11.6 Meherin Sohani

The intelligence service, the continuous integration pipeline, quality assurance and the project documentation.

- The FastAPI intelligence service: photograph classification, severity estimation, relevance checking and image embedding, behind a bounded-timeout interface the platform can run entirely without.
- The model registry, which records the name, version and training status of every model, so that a run of wrong predictions can be traced to the version that produced it.
- The honesty controls on untrained models: a capped confidence ceiling, the citizen category always overriding the model, and every override logged as future training signal.
- The perceptual hash embedding used by deduplication, together with its verification that the transform is orthonormal and invertible.
- The continuous integration pipeline: type-checking both applications, building the client and import-checking the service on every change.
- Project documentation and the contributor guide, including the code-owners mapping that records this division of work inside the repository itself.

How contribution is evidenced

Each member holds commits under their own account in the project repository, and the code-owners file records which directories belong to whom. The division above is therefore verifiable from the repository history rather than asserted only in this document.

12 Appendix: requirement index

GROUP	RANGE	COUNT	SUBJECT
FR-1	FR-1.1 to FR-1.8	8	Account management

GROUP	RANGE	COUNT	SUBJECT
FR-2	FR-2.1 to FR-2.9	9	Problem reporting
FR-3	FR-3.1 to FR-3.8	8	Deduplication
FR-4	FR-4.1 to FR-4.7	7	Priority scoring
FR-5	FR-5.1 to FR-5.7	7	Community verification
FR-6	FR-6.1 to FR-6.10	10	Authority workspace
FR-7	FR-7.1 to FR-7.10	10	Administration
FR-8	FR-8.1 to FR-8.8	8	Public map and dashboard
NFR-1	NFR-1.1 to NFR-1.5	5	Performance
NFR-2	NFR-2.1 to NFR-2.8	8	Security
NFR-3	NFR-3.1 to NFR-3.5	5	Reliability and availability
NFR-4	NFR-4.1 to NFR-4.8	8	Usability and accessibility
NFR-5	NFR-5.1 to NFR-5.5	5	Maintainability and portability
Total		98	67 functional, 31 non-functional

End of document. Amar Shohor, Software Requirements Specification, version 1.0, 23 September 2026.